MTH 200 Chapter 5: Integration

5.5.1 Substitution w/ Indefinite Integrals Worksheet

Use a u-substitution to evaluate each of the following.

1. $\int x \sin\left(x^2\right) dx$

2. $\int (1-2x)^9 \, dx$

3. $\int \sec(3t)\tan(3t)\, dt$

4. $\int (x+1)\sqrt{2x+x^2}\, dx$

5. $\int \sec^2(\theta)\tan^3(\theta)\,d\theta$

6. $\int \frac{\cos(x)}{\sin^2(x)}\,dx$

7. $\int \frac{z^2}{\sqrt[3]{1+z^3}}\,dz$

8. $\int (x^2+1)(x^3+3x)^4\,dx$

9. $\int \sqrt{\cot(x)}\csc^2(x)\,dx$

10. $\int \cos^4(x)\sin(x)\,dx$

11. $\int x(2x+5)^8\,dx$

12. $\int \frac{\cos(\pi / x)}{x^2}\,dx$

13. $\int x^2\sqrt{2+x}\,dx$

14. $\int \sin(t)\sec^2(\cos(t))\,dt$

15. $\int x^3\sqrt{x^2+1}\,dx$

16. $\int \frac{dx}{5-3x}$

17. $\int e^{\tan(x)} \sec^2(x)\,dx$

18. $\int \cot(x)\,dx$

19. $\int e^x \sqrt{1+e^x}$

20. $\int \frac{\sin(x)}{1+\cos^2(x)}\,dx$

21. $\int \frac{1+x}{1+x^2}\,dx$ (Consider using some algebra first before attempting a substitution)